

MSE 360: Kinetic Processes in Materials

Homework #7

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

(There are 2 problems: Problem 2 starts on 2nd page)

1. The following equation (Equation 4.43) provides the solution for transient finite (symmetric) planar diffusion in a plate of thickness L starting from a uniform initial concentration of C_i^0 when the concentrations at the edges of the plate are set to C_i^* at $t=0$:

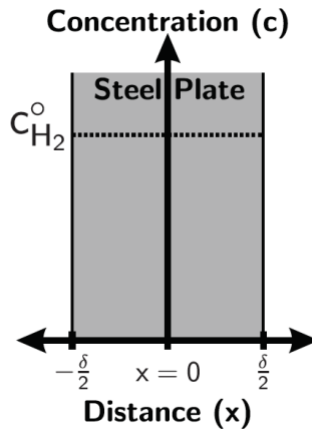
$$C_i(x, t) = C_i^* + (C_i^0 - C_i^*) \frac{4}{\pi} \sum_{j=0}^{\infty} \left(\frac{1}{2j+1} \sin \left[(2j+1) \pi \frac{x}{L} \right] e^{-(2j+1)^2 \pi^2 D t / L^2} \right)$$

At “long times,” this solution can be reasonably approximated by using only the first term in series without significant error. In order to determine a reasonable value for “long time” and to visualize the effect that this approximation has on the solution compared to retaining additional terms in the series, use a software program of your choice (e.g. Excel, Matlab, Mathematica) to make a set of plots for this solution using $j = 1, 3$, and 10 terms in the series at each of the following three times:

- a. $t = 0.01 \left[\frac{L^2}{\pi^2 D} \right]$
- b. $t = 0.1 \left[\frac{L^2}{\pi^2 D} \right]$
- c. $t = 1 \left[\frac{L^2}{\pi^2 D} \right]$

Assume $C_i^* = 0$. Comment on the results.

2. Steel often contains trace amounts of H_2 , which can lead to embrittlement. To avoid embrittlement, steel is often degassed prior to use in order to remove these trace H_2 impurities. Degassing steel involves placing the steel in a vacuum, where the H_2 concentration in the vacuum can be considered zero at all times. Degassing proceeds by three steps: 1) Solid-state diffusion of H_2 from the steel bulk to the steel surface. 2) Desorption of H_2 from the surface of the steel. 3) Gas-phase diffusion of the H_2 away from the steel surface.



Schematic diagram of the cross-section of a steel plate of thickness δ with a constant H_2 concentration of $C_{H_2}^0$.

- Based on what you know about the typical rates of the three steps involved in the steel degasification process, which of the three steps is likely to be the rate-determining step and why?
- Consider a steel plate which initially contains a uniform concentration ($C_{H_2}^0$) of 10^{15} molecules/cm³, as shown above. Assuming that the desorption of H_2 from the surface of the steel is the rate determining step in steel degasification, draw a diagram illustrating the H_2 concentration profile across the steel plate at five times: $t = 0 < t_1 < t_2 < t_3 < t_\infty$.
- Assuming that solid-state diffusion of H_2 from the bulk to the surface of the steel is the rate determining step in steel degasification, draw a diagram illustrating the H_2 concentration profile across the steel plate at five times: $t = 0 < t_1 < t_2 < t_3 < t_\infty$.
- Assuming that solid-state diffusion of H_2 from the bulk to the surface of the steel is the rate determining step in steel degasification, is this a finite, semi-infinite, or infinite boundary-value problem?
- Assuming that solid-state diffusion of H_2 from the bulk to the surface of the steel is the rate determining step in steel degasification, provide the two boundary conditions and one initial condition for this transient diffusion problem. Note that $x = 0$ corresponds to the center of the steel plate. Assume the steel plate is 1 mm thick.

- f) By employing your boundary conditions and initial conditions to evaluate this transient diffusion problem, you arrive at the following solution for the time-dependent concentration of H_2 at the center of the steel plate [$C(x = 0, t)$]:

$$C(0, t) = C_{H_2}^0 \exp \left[-Dt \left(\frac{\pi}{\delta} \right)^2 \right]$$

where δ is the thickness of the steel plate and $C_{H_2}^0$ is the initial ($t = 0$) concentration of H_2 in the steel plate. The H_2 embrittlement is avoided if the H_2 concentration in the steel is reduced below 10^{13} molecules/cm³. If the diffusivity of H_2 in steel under degassing conditions is 10^{-6} cm²/s, how long must the steel be degassed to ensure the H_2 concentration at the center of the plate falls below 10^{13} molecules/cm³? Assume the steel plate is 1 mm thick.

- g) How long must a 4-mm thick steel plate be degassed to avoid H_2 embrittlement?